Coverage-Oriented Planner-Guided Memory Residual Reinforcement Learning for Robotic Thermal-Spot Coverage

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***Abstract*—Localized steam breakthroughs in a granular thermal-processing vessel appear online and remain active until they are covered. We study this persistent service problem in a MuJoCo digital twin of an ABB manipulator, where an external thermal localization module supplies clustered target positions and the robot moves a coverage center over the active spots. A Horizon-2 receding-horizon planner provides a structured action prior, while an LSTM-PPO policy learns a phase-scaled, shielded residual from the active set, spawn history, thermal context, and route summaries. We use three training seeds, three held-out environment seeds, and five episodes per crossed cell. Relative to Horizon-2, coverage is statistically indistinguishable in Low, Realistic, and Hard; in Extreme it increases from 0.816 to 0.855, with a hierarchical 95% interval for the difference of $[0.003, 0.074]$, but the four-stage Holm-adjusted test is not significant. A matched absolute-action ablation that retains the encoder, warm start, prediction loss, and service reward reduces coverage by 0.079–0.168 across the four stages. The Extreme gain is accompanied by higher latency and backlog. The result is therefore a coverage-oriented planner-learning trade-off, not a uniformly faster controller.**

***Index Terms*—robotic coverage, residual reinforcement learning, persistent service, coverage path planning, LSTM-PPO, MuJoCo**

I. Introduction

This work is motivated by robot-assisted loading and steaming of a granular bed. Localized steam breakthroughs are detected by an external thermal sensing system, and an ABB manipulator moves a deposition or coverage center over the vessel to serve each active location. The simulator supplies target coordinates, so the paper evaluates online control and service routing rather than camera perception. Targets arrive in spatially clustered burst–lull cycles, remain active until covered, and can accumulate while the arm is in transit. This makes the task a persistent service problem: geometric travel, waiting time, target age, and congestion must be considered together.

The problem is related to coverage path planning [1], coverage control [2], statistical coverage [3], and persistent monitoring with stochastic arrivals [4]. It also has a direct connection to stochastic dynamic vehicle routing, where online arrivals and waiting time determine service quality [5], [6]. Unlike static map coverage, an active spot leaves the queue only after physical coverage. Reinforcement learning has been applied to adaptive coverage planning [7], [8], but a visible-target planner is already a competitive deterministic prior in this task. Residual reinforcement learning [9] and bounded residual control [10], [11] provide a conservative way to retain that prior. The set encoder follows the permutation-invariant motivation of Set Transformer [12].

The paper is aligned with ICCC Track 6 (Robotics), with secondary links to Tracks 2 (AI for Engineering) and 4 (Machine Learning). We ask: *RQ1*: does a bounded residual action interface improve a fixed planner relative to a matched absolute-action PPO controller? *RQ2*: how does the method compare with alternative planner and heuristic priors? *RQ3*: what coverage–service trade-off is created in latency, strict SLA, backlog, and smoothness?

Learning does not replace planning; it learns a correction around an online route prior. The contributions are (i) a persistent thermal-spot service formulation with explicit metric denominators, (ii) an LSTM-PPO residual interface with phase-scaled authority and a progress shield, and (iii) a crossed held-out evaluation that separates the residual interface from auxiliary modules.

II. Task and Method

A. Persistent service and metrics

Let $\mathcal{S}_t = \{p_i(t)\}$ be the active spot set, where $p_i = (x_i, y_i, a_i)$ contains location and active age, and let $c_t = (x_t, y_t)$ be the coverage center. Spot i is served when

$$\|c_t - (x_i, y_i)\|_2 \leq r_{\text{cover}}. \tag{1}$$

The spot is then removed. Timeout removal is disabled in the reported protocol; an uncleared spot remains in the denominator. The action $u_t = [\Delta x_t, \Delta y_t, s_t] \in [-1, 1]^3$ is issued every

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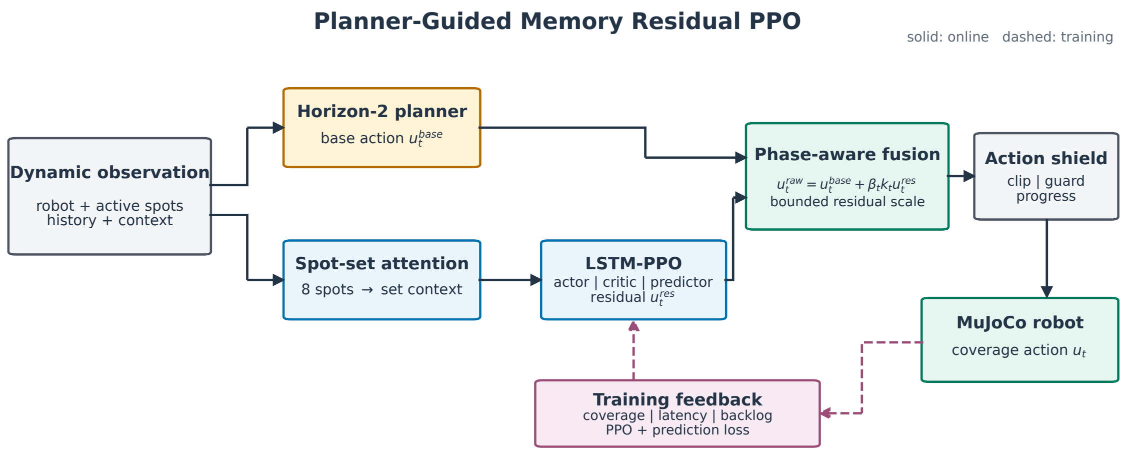

Fig. 1. Planner-guided residual control. The online planner remains in the action loop; LSTM-PPO supplies a bounded correction and the shield checks local progress before the action is sent to the MuJoCo robot.

0.05 s; the MuJoCo integrator uses a 0.002 s physics step. Coverage is

$$C = \frac{N_{\text{covered}}}{N_{\text{spawned}}}. \tag{2}$$

For a covered spot, $T_i = (n_i^{\text{cover}} - n_i^{\text{spawn}}) \times 0.05$ s and $\bar{T} = N_{\text{covered}}^{-1} \sum_i T_i$. Mean latency and p90 are conditional on covered spots; p90 is the empirical 90th percentile of the episode-level covered-spot latency list. Strict SLA exposes the uncleared tail:

$$C_{\text{SLA}} = \frac{|\{i : T_i \le T_{\text{SLA}}\}|}{N_{\text{spawned}}}, \qquad T_{\text{SLA}} = 4.0 \text{ s}. \tag{3}$$

Backlog is the time-average active count; smoothness is $\Delta_u = T^{-1} \sum_t \|u_t - u_{t-1}\|_2$. Metrics are aggregated within evaluation cells before hierarchical resampling, with coverage primary.

B. Planner prior and recurrent residual

The interpretable risk score used by the comparator and as a diagnostic is

$$q_i = 0.40\phi_{\text{age}} + 0.30\phi_{\text{dist}} + 0.10\phi_{\text{reach}} + 0.20\phi_{\text{therm}}. \tag{4}$$

For center c_t, potential radius R_{pot}, home position p_{home}, maximum target offset $R_{\max}$, and stage age scale A_{norm}, the components are

$$\begin{aligned} \phi_{\text{age}} &= \text{clip}(a_i/A_{\text{norm}}, 0, 1), \\ \phi_{\text{dist}} &= 1 - \text{clip}(\|p_i - c_t\|/R_{\text{pot}}, 0, 1), \\ \phi_{\text{reach}} &= 1 - \text{clip}(\|p_i - p_{\text{home}}\|/R_{\max}, 0, 1), \\ \phi_{\text{therm}} &= \text{clip}(H(p_i), 0, 1). \end{aligned} \tag{5}$$

The implementation also exposes $\phi_{\text{mat}} = \text{clip}(\text{mean}_g[\max(H^* - H_g, 0)\exp(-\|x_g - p_i\|^2/(2\sigma_d^2))]/H^*, 0, 1)$. Here x_g is the center of grid cell g, H_g is its height, H^* is the target layer height, and $\sigma_d = 0.18$ m is the deposition-kernel width; the mean is taken over the grid. The thermal score is $H(p) = \text{clip}(\sum_\ell a_\ell \exp(-\|p - h_\ell\|^2/(2\sigma_h^2)), 0, 1)$, where h_ℓ and a_ℓ are hotspot centers and amplitudes and $\sigma_h = 0.22$ m. V12 disables urgency augmentation, so it does not enter either q_i or the H2 objective. When material observation is enabled, the weights are $(.35, .25, .15, .10, .15)$ for age, distance, material, reachability, and thermal context; when it is disabled, they are $(.40, .30, 0, .10, .20)$. The reported V12 protocol sets `material_observation=false`, hence $\phi_{\text{mat}} = 0$ and its weight is zero. The risk-aware baseline is stateless: it selects the first active-set entry attaining the maximum score and outputs the normalized direction toward that target with speed one.

Horizon-L enumerates ordered routes of length $\min(L, N_t)$. For route $\rho = (i_1, \dots, i_L)$, the planner simulates travel in steps and uses

$$\begin{aligned} g_j &= q_{i_j} + 0.45 z_j - 0.018 d_j - 0.050 (z_j - 1)_+, \\ J(\rho) &= \sum_{j=1}^{L} 0.82^{j-1} g_j - 0.08 \bar{z}_{\text{left}}, \end{aligned} \tag{6}$$

where d_j is travel steps and z_j is arrival age normalized by A_{norm}; the simulated center and elapsed travel time are updated after each route element. H2 uses $L = 2$ and executes the first action of the highest-scoring route; H3 uses the identical objective with $L = 3$. The planner enumerates all ordered routes from the active set, uses $d_j = \max(1, \lceil \max(\|p_{i_j} - c\| - r_{\text{cover}}, 0)/0.04 \rceil)$, and retains the first enumerated route under an exact tie. The resulting action is the normalized direction to the first route target with speed one, clipped to $[-1, 1]^3$. Before execution, the environment applies action smoothing $\bar{u}_t = 0.7\bar{u}_{t-1} + 0.3\tilde{u}_t$. For raw action $u_t^{\text{raw}} = [d_x, d_y, d_s]$, the planar displacement is $0.04\,\eta_t [d_x, d_y]$, where $\eta_t = 0.65 + 0.35(d_s + 1)/2$. Nearest, oldest, distance–age, dynamic-weighted, ACO-TSP, and planner-ensemble baselines

TABLE I
HORIZON-2 TARGET-SELECTION PROCEDURE

Input: active set $\mathcal{S}_t$, center c_t, and $L = 2$.
1. Enumerate $\mathcal{R}_t = \mathrm{Perm}(\mathcal{S}_t, \min(2, |\mathcal{S}_t|))$ and evaluate $J(\rho)$ for each route.
2. Set $\rho^* = \arg\max J(\rho)$; strict $>$ preserves enumeration order on ties.
3. Output the normalized direction to $p_{\rho_1^*}$ with speed one, clipped to $[-1, 1]^3$.
4. Recompute at the next decision step; no planner hidden state is carried.

use the same workspace, action interface, persistence rule, and coverage condition.

The observation has $61 + 8 \times 8 = 125$ features. The global branch is $61 \to 256$ with LayerNorm and ReLU. Eight spot slots are embedded with a shared $8 \to 128$ map and four-head masked attention; the pooled spot branch is $128 \to 256$ with LayerNorm and ReLU. Their concatenation is fused by $512 \to 256 \to 256$ LayerNorm–ReLU blocks and fed to a one-layer width-256 LSTM. The fused representation is processed by $256 \to 3$ actor, $256 \to 1$ critic, and $256 \to 128 \to 2$ prediction heads. Actor means use tanh and the state-independent log standard deviation is initialized to -0.8. Relative target and active-spot coordinates and distances are divided by the pot radius; ages are divided by A_{norm}; thermal, material, risk, coverage, route, and spawn-readiness features use bounded physical scales. Joint coordinates and end-effector tracking errors remain in simulator units, and actions are bounded in $[-1, 1]$. No running test-set observation normalizer is used. The hidden state is carried across rollout chunks and coverage events, reset after a miss and at an episode boundary. With planner action u_t^{base} and residual $u_t^{\mathrm{res}} = \pi_\theta(o_t, h_t)$, the candidate is

$$
\begin{aligned}
\tilde{u}_t &= \mathrm{clip}(u_t^{\mathrm{base}} + \beta_t k_t u_t^{\mathrm{res}}, -1, 1), \\
\beta_t &= 0.03 + 0.17\,\mathrm{clip}(n_t/700000, 0, 1).
\end{aligned} \tag{7}
$$

The phase multipliers (k_t) for lull, sparse, charging, mid, dense, and burst are $(1.35, 1.25, 1.00, 1.00, 0.40, 0.25)$. The post-policy shield rejects alignment below 0.15, requires candidate progress of at least 0.35 of base progress with margin 0.002, and uses a 0.70-base/0.30-candidate guarded blend with a 0.50 progress ratio. Reverse turns below cosine -0.55 are blended, and a candidate worse than the base is rejected after 90 no-cover steps. It is a local progress filter, not a formal temporal-logic safety guarantee [13].

The scalarized reward contains time, active count, mean age, potential, best progress, coverage, quick cover, precision, latency, oldest active, backlog, SLA, action change, action norm, miss, and success terms. The stage base cover, quick-cover, and potential values are $(68, 64, 62, 60)$, $(16, 14, 12, 10)$, and $(8.0, 7.5, 6.5, 5.8)$; the V12 scale factors are $(.70, 3.0, .65, .65, .45, 2.4, 2.8)$ for cover, quick cover, precision, potential, best progress, active count, and age. Thus the corresponding scaled coefficients are cover $(47.6, 44.8, 43.4, 42.0)$, quick cover $(48, 42, 36, 30)$, precision 7.8, potential $(5.20, 4.875, 4.225, 3.77)$, best progress .3375, active count .024, and age .112. The remaining coefficients are $r_{\mathrm{time}} = (.030, .035, .045, .055)$, $r_{\mathrm{latency}} = 30$, $r_{\mathrm{oldest}} = .45$, $r_{\mathrm{backlog}} = .20$, $r_{\mathrm{SLA}} = (20, -30)$, $r_{\Delta u} = .025$, $r_{\|u\|} = .004$, $r_{\mathrm{miss}} = 28$, and $r_{\mathrm{success}} = 30$. The timeout-miss term is inactive because timeout removal is disabled. The reward is multiplied by 0.02 and clipped to $[-4, 4]$; SLA and backlog are therefore scalar preferences, not hard constraints. The success bonus is emitted when covered count reaches the stage target and coverage reaches the configured target coverage.

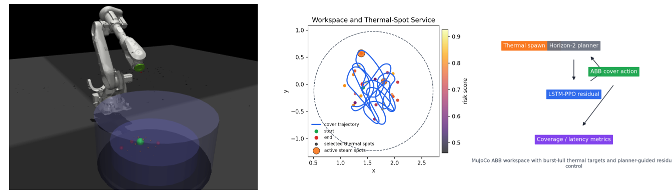

Fig. 2. MuJoCo ABB workspace (left) and online thermal-spot service task (right). The controller moves the green coverage center over clustered spots inside the circular vessel.

III. Experiments

A. Protocol and statistical unit

The MuJoCo twin [14] contains an ABB arm, circular vessel, and external target field. Three drifting hotspots generate a 70-step lull, 110-step charge, then 3–5 spots emitted every four steps; background and strength are 0.055 and 3.8. Each learned method uses train seeds 0–2, held-out seeds 100–102, five 3,200-decision-step episodes per crossed cell, and the terminal checkpoint after a fixed curriculum. The training budget is $1,152 \times 800 = 921,600$ decision steps per seed. PPO uses $(\gamma, \lambda) = (.985, .95)$, Adam with 10^{-5} learning rate and $\epsilon = 10^{-5}$, one epoch, clips $(.05, .05)$, value coefficient .5, gradient norm .35, entropy $8 \times 10^{-4} \to 1.8 \times 10^{-4}$ over 10^6 steps, curriculum 80/160/720/192 episodes from Low to Extreme, H2 BC for 200 episodes (20/40/100/40), 12 BC epochs, prediction coefficient .05 with a 150-step horizon, BC decay $0.03 \to 0$ over 260,000 steps, and residual schedule $0.03 \to 0.20$ over 700,000 steps. The PPO objective additionally uses action-smooth and action-ℓ_2 coefficients $(.02, .002)$; advantages are standardized within each update. The budget therefore contains $921,600/3,200 = 288$ PPO updates. Checkpoints are written every 50 training episodes and at curriculum-stage boundaries; the reported file is the terminal `scheme_d_paper_base_latest_full.pt` written after episode 1,152. Held-out data do not select checkpoints. Matched component variants use the same fixed V12 settings, budget, and action interface; no common hyperparameter search was completed. The historical Vanilla diagnostic was not independently retuned under a full-budget protocol. The archived ‘episodes.csv‘ files record reward, coverage, SLA, residual scale, recurrent counters, prediction events, and elapsed steps.

The reported evaluation uses a fixed 3,200-decision-step service window, equivalent to 160 s at 0.05 s per decision. Timeout removal and terminal drain-to-clear are disabled in this protocol; an active but uncovered spot remains in the denominator and contributes to the pending backlog at the window boundary.

Each episode first yields coverage, latency, SLA, backlog, and smoothness values; p90 is computed from that episode’s

TABLE II
STAGE AND TRAINING PARAMETERS

Stage	$N_{\max}$	r_{cover}	A_{norm}	p_{spawn}	cooldown
Low	3	.17	260	.006	80
Realistic	4	.14	220	.010	65
Hard	6	.12	200	.018	48
Extreme	8	.10	180	.026	36
Obs. / widths	125; spot 128, global/LSTM 256; 4-head attention				
Sequence / update	$4 \times 800 = 3200$; PPO update every 4 chunks				

TABLE III
COVERAGE VERSUS HORIZON-2 (HIERARCHICAL 95% CIS)

Stage	Ours	H2	Δ	Δ CI	p_{Holm}
Low	.917 [.906,.929]	.919 [.906,.932]	-.002	[-.018,.016]	1.000
Real.	.901 [.884,.922]	.896 [.883,.910]	+.005	[-.018,.034]	1.000
Hard	.870 [.844,.894]	.870 [.847,.894]	$<$ .001	[-.040,.033]	1.000
Extreme	.855 [.828,.881]	.816 [.782,.849]	+.038	[.003,.074]	.135

covered-spot latency list and is not recomputed by pooling spots across episodes. Cell summaries average the five episodes, and the balanced crossed design is then summarized over training and held-out environment seeds. We use 20,000 crossed bootstrap draws: training labels, evaluation labels, and episodes are resampled at their respective levels, while paired methods reuse the same held-out scenarios. Deterministic planners omit the training-seed level. Tables report hierarchical 95% intervals; Holm correction is applied across the four stages for each predeclared comparison family. This keeps training randomness and environment randomness from being treated as 45 independent flat observations.

B. Coverage and baselines

Table III is the primary comparison. Ours is not distinguishable from H2 in the first three stages; the Extreme interval is positive before, but not after, four-stage Holm correction. Risk-aware has a higher point estimate in Extreme, while H3 and the planner ensemble have comparable point estimates in selected stages. Its non-monotonic coverage $(.924, .901, .836, .863)$ follows from jointly changing capacity, radius, arrival rate, and A_{norm}; it has no hidden state.

C. Action-interface and component evidence

The matched No-residual policy retains the observation, LSTM, H2 warm start, prediction loss, service reward, and budget, but emits an absolute action without a planner base or shield. This is the controlled action-interface comparison. Historical Vanilla LSTM-PPO changes residual composition, BC, and prediction jointly, and was not independently retuned under a full-budget protocol; it is descriptive rather than causal. Table V(b) records the differences.

D. Service diagnostics and robustness

Table V(c) reports service cost. In Extreme, the 0.038 coverage increase has 2.50 s higher mean latency, 6.09 s higher p90, 0.26 more active spots, and lower strict SLA. On an unseen higher-density condition, ours obtains .868/.850 in Hard/Extreme and H2 .888/.825; overlapping intervals make

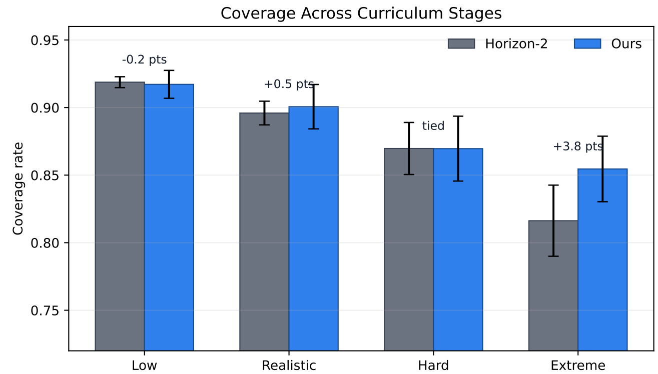

Fig. 3. Stage-wise coverage with hierarchical 95% intervals. The separation is concentrated in Extreme and does not remain significant after correction.

TABLE IV
COVERAGE ACROSS DETERMINISTIC AND LEARNED BASELINES. BRACKETS ARE HIERARCHICAL 95% CIS; ± VALUES ARE EVALUATION-SEED STANDARD DEVIATIONS.

Method	Low	Real.	Hard	Extreme
Ours	.917 [.906,.929]	.901 [.884,.922]	.870 [.844,.894]	.855 [.828,.881]
Horizon-2	.919 [.906,.932]	.896 [.883,.910]	.870 [.847,.894]	.816 [.782,.849]
Horizon-3	.917±.009	.927±.025	.860±.022	.841±.060
Risk-aware	.924 [.902,.953]	.901 [.885,.918]	.836 [.778,.876]	.863 [.833,.892]
Nearest	.922±.011	.887±.016	.860±.026	.789±.050
Oldest	.914±.004	.890±.014	.883±.019	.807±.032
Distance–age	.921±.016	.896±.008	.851±.020	.801±.065
Dynamic weighted	.927±.003	.886±.023	.865±.005	.799±.045
ACO-TSP	.925±.006	.893±.004	.864±.033	.754±.061
Planner ensemble	.935±.012	.899±.014	.870±.003	.814±.090
No residual	.838 [.712,.915]	.776 [.667,.865]	.726 [.590,.837]	.687 [.585,.793]
Vanilla LSTM-PPO	.166 [.067,.286]	.130 [.048,.234]	.089 [.025,.181]	.061 [.017,.115]

The deterministic rows use the same environment seeds and action interface; Horizon-L uses the objective defined above. Vanilla is reported as a descriptive diagnostic because its training protocol is confounded, not as a causal baseline.

this robustness evidence. H3 tests planner capacity only; matched H3-residual retraining is not claimed.

IV. DISCUSSION AND CONCLUSION

The matched action-interface comparison is consistent with a contribution from the planner-residual interface; it does not establish dominance over H2 or the risk-aware heuristic. H2 is not distinguishable from the learned method in three stages, while the Extreme difference is positive before but not after Holm correction. The historical Vanilla result is confounded by simultaneous changes to residual composition, BC, and prediction, and is therefore diagnostic only. The auxiliary ablations do not establish independent gains for attention, carry, prediction, or service shaping.

H3 evaluates deterministic planner capacity, not residual transfer, because the learned controller was trained around H2. Finally, the 0.05 s interval measures task time rather than inference time, and the fixed window does not enforce terminal clearance. The evidence therefore concerns fixed-horizon persistent service; terminal-drain evaluation, matched H3-residual training, and hardware timing remain future work.

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TABLE V

MECHANISM AND SERVICE EVIDENCE: (A) MATCHED RESIDUAL-INTERFACE EFFECT, (B) COMPONENT ACCOUNTING, AND (C) COVERAGE–SERVICE DIAGNOSTICS.

(a) Matched residual-interface effect

Stage	Full	No residual	Δ	Δ CI	p_{Holm}
Low	.917	.838 [.712,.915]	+.079	[.004,.202]	.029
Real.	.901	.776 [.667,.865]	+.124	[.036,.223]	.003
Hard	.870	.726 [.590,.837]	+.144	[.037,.268]	.011
Extreme	.855	.687 [.585,.793]	+.168	[.050,.285]	.003

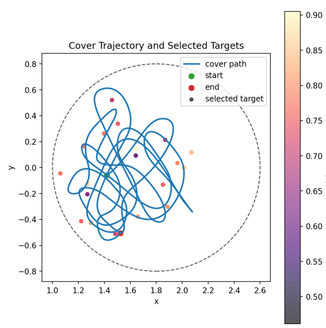
(b) Exact component accounting

Variant	Action	BC	Attn.	Carry	Pred.	Svc.	Shield
Full	H2+res.	Y	Y	Y	Y	Y	Y
No residual	absolute	Y	Y	Y	Y	Y	N
No attention	H2+res.	Y	N	Y	Y	Y	Y
No carry	H2+res.	Y	Y	N	Y	Y	Y
No prediction	H2+res.	Y	Y	Y	N	Y	Y
No service	H2+res.	Y	Y	Y	Y	N	Y
Vanilla	absolute	N	Y	Y	N	Y	N

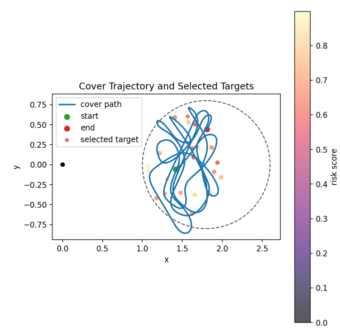
(c) Coverage and service diagnostics versus H2

Stage	Method	Cov.	Mean lat. (s)	p90 (s)	Strict SLA	Backlog	Δ_u
Low	Ours	.917	12.54	24.15	.181	2.30	.0254
	H2	.919	12.16	23.72	.191	2.28	.0255
Real.	Ours	.901	16.04	29.70	.147	2.84	.0261
	H2	.896	16.52	29.45	.131	2.82	.0258
Hard	Ours	.870	19.59	38.10	.127	3.49	.0277
	H2	.870	19.23	35.44	.121	3.43	.0274
Extreme	Ours	.855	23.88	46.01	.103	3.81	.0289
	H2	.816	21.38	39.92	.128	3.55	.0277

Mean and p90 latency condition on covered spots; strict SLA uses all spawned spots. Backlog is mean active count and Δ_u is mean per-step action change.



(a) Planner-guided residual PPO



(b) Horizon-2

Fig. 4. Representative held-out Realistic trajectories. Aggregate claims use the crossed training/evaluation seed cells.

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